

Task ‘Understanding Cloud Architecture and Implementing Different Frameworks’ Unit 2

Formative Discussion ROCCA and ToGAF

The study by Anggraini, Binariswanto and Legowo (2019) presents a practical approach for government agencies seeking to adopt cloud computing by integrating the ROCCA model with the TOGAF 9.2 enterprise architecture framework. Rather than viewing cloud adoption as a purely technical initiative, the authors demonstrate that successful implementation requires alignment between organizational strategy, processes, governance, business requirements and technology architecture.

(Anggraini, Binariswanto and Legowo, 2019).

The mapping of ROCCA processes to TOGAF 9.2 phases provides a structured methodology that agencies can utilize and be able to evaluate business needs, define cloud strategies and develop implementation roadmaps that are very aligned with business vision and targets.

A key strength of both frameworks is its emphasis on strategic planning before technology deployment. The case study of SKK Migas which operates in the energy branch, and which was designated by the authorities, illustrates how critical infrastructure and government institutions can address operational challenges, such as increasing data growth and limited infrastructure capacity, while ensuring that cloud adoption is fully aligned with organizational objectives. The authors also acknowledge that issues including security, data protection

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and elements must be incorporated into the governance and planning process rather than treated as secondary considerations.

(Anggraini, Binariswanto and Legowo, 2019).

Altogether, the combined use of ROCCA and TOGAF 9.2 offers government agencies a well-structured and business-driven approach to cloud transition that supports effective business alignment and long-term digital transformation.

Reference

- Anggraini, N., Binariswanto and Legowo, N. (2019) Cloud Computing Adoption Strategic Planning Using ROCCA and TOGAF 9.2: A Study in Government Agency. *Procedia Computer Science*, 161, pp. 1316–1324. Available at: https://www.researchgate.net/publication/338353469_Cloud_Computing_Adoption_Strategic_Planning_Using_ROCCA_and_TOGAF_92_A_Study_in_Government_Agency [Accessed: 2 August 2026].

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